UNIT 1 · 8 hours

Mechanical Rotational Systems and Gear Trains



Introduction and Mathematical Modelling of Physical Systems

Control Systems (22EEG53)
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U1 Learning objectives

- List rotational elements and their torque relations
- Derive transfer functions for rotational systems
- Reflect inertia, friction and torque through gear trains
- Obtain ωn and ζ of a rotational system

Syllabus context

Introduction and mathematical modelling of physical systems: open loop and closed loop systems, examples, control system components. Transfer function of physical systems: mechanical systems (translational and rotational), electrical systems, analogous systems, block diagram reduction techniques, signal flow graphs.

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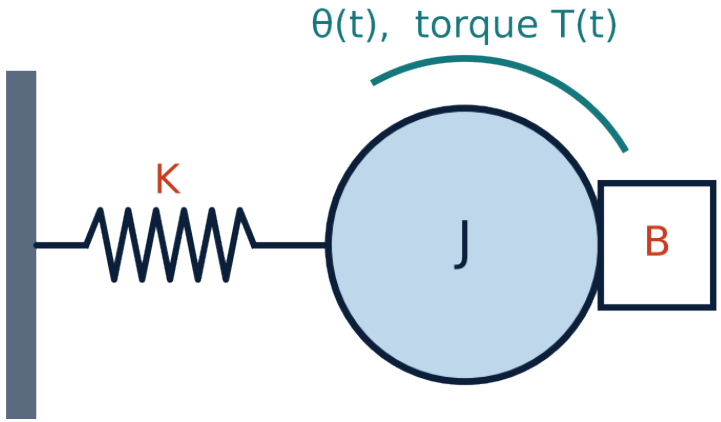
Rotational elements

Element	Torque relation	Laplace
Moment of inertia J	$T = J\, d^2\theta/dt^2$	$J\, s^2\, \Theta(s)$
Viscous friction B	$T = B\, d\theta/dt$	$B\, s\, \Theta(s)$
Torsional spring K	$T = K\, \theta$	$K\, \Theta(s)$

Same structure as translational elements with T ↔ f and θ ↔ x.

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Rotational system



- Disc of inertia J on a shaft with torsional spring K and viscous friction B.
- Applied torque T(t).
- $J\ \ddot{\theta} + B\ \dot{\theta} + K\ \theta = T(t)$.
- $\Theta(s)/T(s) = 1/(J\ s^2 + B\ s + K)$.

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Parameters

$$\Theta(s)/T(s) = 1/(J\,s^2 + B\,s + K)$$

transfer function

$$\omega n = \sqrt{(K/J)}$$

natural frequency

$$\zeta = B / (2\sqrt{(KJ)})$$

damping ratio

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Gear trains

Gear ratio

$N_1 \theta_1 = N_2 \theta_2$, so $\theta_2/\theta_1 = N_1/N_2$ and $T_2/T_1 = N_2/N_1$.

Reflected inertia

Referred to shaft 1: $J_{eq} = J_1 + (N_1/N_2)^2 J_2$.

Reflected friction

$B_{eq} = B_1 + (N_1/N_2)^2 B_2$ (same factor).

Parameters are referred to the shaft where the equation is written, multiplied by $(N_1/N_2)^2$ when moved from the load to the motor side.

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Numerical: rotational system

Problem $J = 0.5$ kg·m², $B = 0.2$ N·m·s/rad, $K = 2$ N·m/rad. Find Θ/T, ωn, ζ and classify.

- $\Theta/T = 1/(0.5\,s^2 + 0.2\,s + 2) = 2/(s^2 + 0.4\,s + 4)$
- $\omega n = \sqrt{(K/J)} = \sqrt{4} = 2$ rad/s
- $\zeta = B/(2\sqrt{(KJ)}) = 0.2/(2 \times 1) = 0.1$

Answer: $\omega n = 2$ rad/s, $\zeta = 0.1$: lightly damped (underdamped)

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Numerical: gear train

Problem Motor shaft J1 = 0.01, gear ratio N1/N2 = 1/10 (load turns slower), load J2 = 4 kg·m². Find the equivalent inertia at the motor shaft.

- $(N1/N2)^2 = 0.01$
- $J_eq = J1 + 0.01 \times J2 = 0.01 + 0.04$

Answer: J_eq = 0.05 kg·m² at the motor shaft

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Key takeaways

- Rotational: $T = J\ddot{\theta} + B\dot{\theta} + K\theta$
- $\Theta/T = 1/(Js^2 + Bs + K)$; $\omega_n = \sqrt{(K/J)}$, $\zeta = B/(2\sqrt{(KJ)})$
- Gear trains: $\theta_2/\theta_1 = N_1/N_2$, $T_2/T_1 = N_2/N_1$
- Reflect parameters with $(N_1/N_2)^2$

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Practice questions

Q1. Derive Θ(s)/T(s) for a disc with spring and damper.

Q2. Find J_eq for J1 = 0.02, J2 = 2, N1/N2 = 1/5.

Q3. Why do we reflect inertia through gears?

Q4. J = 1, B = 4, K = 4: find ζ and ωn.